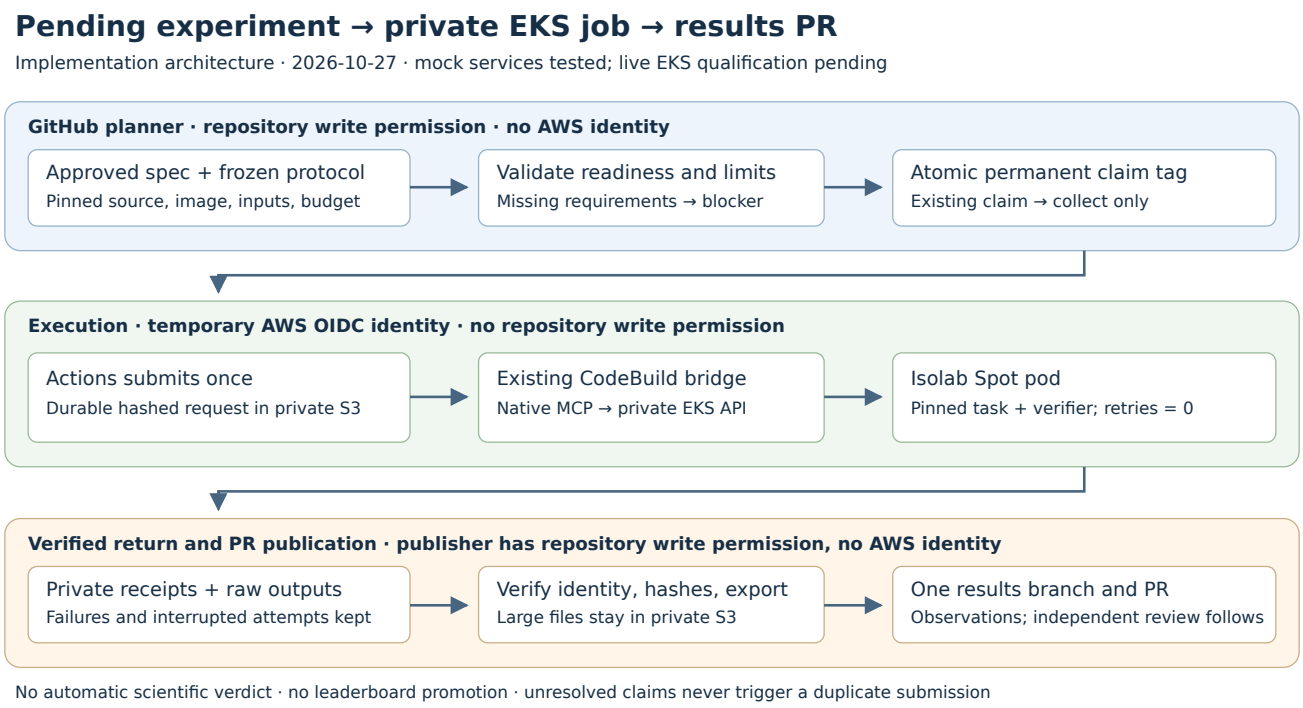
Automatic Isolab experiment queue

Implementation and validation · 2026-10-27

The requested feature discovers reviewed, unrun experiments, submits bounded jobs to the existing Isolab EKS cluster, and returns observations in a results PR. The controller, GitHub workflow, two execution adapters, and recovery path are implemented. Live cloud qualification remains outstanding.



Dispatch. A registration binds an approved expspec record, prospectively frozen protocol, immutable source and image, input hashes, independent verifier, and resource envelope. An atomic GitHub tag prevents duplicate execution. Existing claims only collect. CodeBuild provides the private-EKS route using native MCP. The Actions runner executes no cryptographic workload.

Return. Job/Pod identity, observed image, request and output hashes, source, protocol, export schema, and execution-cell coverage are checked. Native benchmark L2, replay, input-spec, and A/A admission are retained. Compact public observations are committed on a results branch; raw data stays in private S3. Failure and invalid records remain observations pending independent scientific review.

Validation. The native test suite uses fake GitHub/AWS/MCP services and sealed fixture observations. It exercises concurrent claims, uncertain submission, restart collection, changed identities, literal argv, independent verifier failure, canonical export compatibility, scoped PR publication, and queue progress. Exact commands, counts, and failures are in `validation.json` beside this report. Tests do not establish live EKS or IAM readiness.

Deployment gaps. This session has no AWS identity; no real EKS job was launched. The workflow must land on main with OIDC/project/bucket variables and a private canonical-schema reader token configured. Hosted runner billing must permit job starts. The NIST filtering entry remains blocked on published checkpoint inputs, a tested task image, a prospective EKS and storage/budget amendment, and approval. Resumable/CRIU protocols are not adapted by this ordinary-job queue.

Sources. Repository `queue-experiment`, `isolab-eks`, native expspec and remote build verifier; canonical protocol/export schemas at crypto-autoresearcher commit `6a69b2d02ecd53f1649bcf56e23e4ffe91d953ad`; AWS CodeBuild StartBuild; GitHub Git references and workflow token event rules. The setup guide has links, field requirements, credential boundaries, and recovery instructions.